



First Name: _____

Family Name: _____

Student Number: _____

Diploma of Engineering
University Entrance Program
Foundation Studies Program

DPEN021 – Enabling Mathematics A

Skill Mastery Exam 1: Sample

This exam represents 24% of the total marks for the subject

Time allowed: 40 minutes
Items permitted in exam: UOW approved calculator

INSTRUCTIONS TO STUDENTS

1. Write in blue or black pens only.
2. All questions are to be attempted.
3. All notation is as given in lectures.
4. Show all necessary WORKING and reasoning in the spaces provided.
5. For full marks to be awarded, your working and reasoning need to be consistent with your selected answer. A correct answer with no working may receive a mark of zero.
6. You must achieve 80% on a Skill Mastery Exam 1 to be eligible to sit other exams in DPEN021.
7. Total number of questions: 12
 - All questions are multiple choice
 - Each question is worth 2 marks.
8. On the answer sheet provided, circle the letter (A, B, C, D, or E) corresponding to the correct answer.

No examination material may be removed from the examination venue.

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Show all necessary WORKING and reasoning in the spaces provided. On the answer sheet provided, circle the letter (A, B, C, D, or E) corresponding to the correct answer.

Question 1 [Expansion]

The expression $3x(x-1) - 2(x^2 + 2x + 5)$ can be simplified as:

- (A) $x^2 + x - 10$
- (B) $5x^2 + 7x - 10$
- (C) $x^2 - 7x - 10$
- (D) $-7x - 10$
- (E) None of the above

Question 2 [Expansion]

Expand and simplify $4(a-3b)^2 - (a^2 - 2)$.

- (A) $3a^2 - 24ab + 36b^2 + 2$
- (B) $4a^2 - 24ab + 36b^2 - 2$
- (C) $3a^2 - 6ab + 9b^2 + 2$
- (D) $3a^2 - 24ab + 9b^2 - 2$
- (E) $a^2 - 24ab + 36b^2 + 2$

Question 3 [Pascal's Triangle]

What is the sum of all the coefficients in the Pascals Triangle used to expand $(a + b)^5$?

- (A) 5
- (B) 10
- (C) 16
- (D) 32
- (E) 64

Question 4: [Pascal's Triangle] (MATH141 questions)

Using Pascal's Triangle, the expansion for $(2y - x)^4$ is:

- (A) $16y^4 + 32y^3x + 24y^2x^2 + 8yx^3 + x^4$
- (B) $16y^4 - 8y^3x + 24y^2x^2 - 8yx^3 + x^4$
- (C) $16y^4 - 32y^3x + 24y^2x^2 - 8yx^3 + x^4$
- (D) $8y^4 - 32y^3x + 24y^2x^2 - 8yx^3 + x^4$
- (E) $16y^4 - 32y^3x + 12y^2x^2 - 8yx^3 + x^4$

Question 5 [Factorising]

Expand and then factorise: $2(4a^2 + 2) - 6a^2 - 12$

- (A) $2(a - 2)(a + 2)$
- (B) $8 - 2a^2$
- (C) $14a^2 - 20$
- (D) $8 + 2a^2$
- (E) $2a(a - 4)$

Question 6 [Factorise]

Factorise $3ac - 2da + 3bc - 2bd$

- (A) $(a - b)(2d - 3c)$
- (B) $(a + b)(2d - 3c)$
- (C) $(a + b)(3c - 2d)$
- (D) $(a - b)(3c - 2d)$
- (E) $(a + b)(3c + 2d)$

Question 7 [Factorising] (MATH141 question)

Factorise $(3x + 4y)^2 - (2x - y)^2$.

- (A) $(x + 5y)(5x + 3y)$
- (B) $(x + 3y)(5x + 5y)$
- (C) $(3x + 4y)(2x - y)$
- (D) $(x - 5y)(5x + 3y)$
- (E) $(5x + y)(x + 3y)$

Question 8 [Factorise Quadratic or Cubic]

Factorise completely $24 + 3x^3$.

- (A) $3(8 + x^3)$
- (B) $3(2 + x)(4 + x^2)$
- (C) $3(2 + x)(4 - 2x + x^2)$
- (D) $3(2 + x)(4 + 2x - x^2)$
- (E) None of the above

OR

Factorise $2x^2 - 5x - 12$.

- (A) $(2x - 3)(x + 4)$
- (B) $(2x + 3)(x - 4)$
- (C) $(2x + 2)(x - 6)$
- (D) $(2x - 4)(x + 3)$
- (E) $(2x + 3)(x - 4)$

Question 9 [Algebraic Fractions]

Simplify the following fractions $\frac{x^4 - x^3 + 2x - 2}{x^2 + 2x - 3}$.

- (A) $\frac{x^3 + 2}{x - 1 + 3}$
- (B) $\frac{x^3 + 2}{x + 3}$
- (C) $\frac{x^4 + 2}{x^2 + 2x - 3}$
- (D) $x^3 + 2$
- (E) $\frac{x^3 - 2}{x + 3}$

Question 10 [Algebraic Fractions – adding/subtracting/division/multiplication]

Simplify $\frac{3}{x^2 - 4} - \frac{2}{x^2 - 3x + 2}$.

- (A) $\frac{1}{(x^2 - 4)(x^2 - 3x + 2)}$
- (B) $\frac{5x + 1}{(x^2 - 4)(x^2 - 3x + 2)}$
- (C) $\frac{x - 7}{(x - 2)(x + 2)(x - 1)}$
- (D) $\frac{5x + 1}{(x^2 - 4)(x - 1)}$
- (E) $\frac{1}{-2 - 3x}$

Question 11 [Surd]

Simplify $12\sqrt{48} + \sqrt{343} - 3\sqrt{112} - 20\sqrt{3}$.

- (A) $28\sqrt{3} - 5\sqrt{7}$
- (B) $5\sqrt{204}$
- (C) 56
- (D) $7\sqrt{3}$
- (E) $7\sqrt{3} - \sqrt{7}$

Question12: [Surd]

Rationalise the denominator and simplify: $\frac{1}{\sqrt{3} + 2\sqrt{5}}$

- (A) $\frac{2\sqrt{5} + \sqrt{3}}{-17}$
- (B) $\frac{2\sqrt{5} - \sqrt{3}}{-17}$
- (C) $\frac{\sqrt{3} - 2\sqrt{5}}{17}$
- (D) $\frac{2\sqrt{5} - \sqrt{3}}{17}$
- (E) $\frac{2\sqrt{5} + \sqrt{3}}{17}$

END OF EXAM.

Formula Sheet

Expansion

Distributive Law:

$$a(b \pm c) = ab \pm ac$$

Distributive Law Twice:

$$\begin{aligned}(a + b)(c + d) &= a(c + d) + b(c + d) \\ &= ac + ad + bc + bd\end{aligned}$$

Pascal Triangle can be used for binomial expansions of the form $(a \pm b)^n$

Square of Binomial:

$$\begin{aligned}(a + b)^2 &= a^2 + 2ab + b^2 \\ (a - b)^2 &= a^2 - 2ab + b^2\end{aligned}$$

Expansion resulting in Difference of Two Squares

$$(a - b)(a + b) = a^2 - b^2$$

Factorisation

Distributive Law:

$$ab \pm ac = a(b \pm c)$$

Compound Common Factor:

$$\begin{aligned}ax + bx + ay + by &= x(a + b) + y(a + b) \\ &= (a + b)(x + y)\end{aligned}$$

Difference of Two Squares:

$$a^2 - b^2 = (a - b)(a + b)$$

Sum and Differences of Two Cubes:

$$\begin{aligned}a^3 + b^3 &= (a + b)(a^2 - ab + b^2) \\ a^3 - b^3 &= (a - b)(a^2 + ab + b^2)\end{aligned}$$

Quadratic Factors:

$$x^2 + (a + b)x + ab = (x + a)(x + b)$$

Rules for Surds

Multiplication of Surds:

$$\sqrt{xy} = \sqrt{x} \times \sqrt{y}$$

Division of Surds:

$$\sqrt{\frac{x}{y}} = \frac{\sqrt{x}}{\sqrt{y}}$$

Like surds can be added and subtracted

To rationalise a binomial denominator, use the result:

$$(\sqrt{a} - \sqrt{b})(\sqrt{a} + \sqrt{b}) = a - b$$



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Skill Mastery Exam 1 Answer Sheet: Sample

This exam represents 12% of the total marks for the subject

Circle the letter (A, B, C, D, or E) corresponding to the correct answer.

1. A B C D E
2. A B C D E
3. A B C D E
4. A B C D E
5. A B C D E
6. A B C D E
7. A B C D E
8. A B C D E
9. A B C D E
10. A B C D E
11. A B C D E
12. A B C D E